

Unit 6: Internet Security

What you need to know: This unit focuses on the various digital threats that can harm your computer or compromise your personal data, as well as how to protect yourself against them.

- **Malware**

- **What it is:** A broad umbrella term that stands for "Malicious Software."
- **Study Tip:** Remember that viruses, Trojans, worms, and spyware are all just different specific *types* of malware. If it's software designed to do bad things, it is classified as malware.

- **Virus**

- **What it is:** A malicious program that attaches itself to a clean file or program and spreads to other files when that infected program is executed.
- **Study Tip:** Just like a biological virus needs a host cell to spread, a computer virus needs a "host" file. It almost always requires a human to actively open, run, or share the infected file to cause damage.

- **Trojan**

- **What it is:** Named after the wooden "Trojan Horse" from Greek mythology, this is malware intentionally disguised as legitimate, harmless software.
- **Study Tip:** You might download what you think is a free game or a helpful tool, but once you install it, it secretly drops a malicious payload or opens a "backdoor" for hackers to access your system. Unlike viruses, Trojans do not copy or replicate themselves.

- **Worm**

- **What it is:** A highly dangerous standalone program that replicates itself incredibly fast and spreads rapidly across computer networks.

- **Study Tip:** The key difference between a Worm and a Virus is that a worm does *not* need human interaction (like clicking a file) to spread. It actively crawls through internet and network vulnerabilities entirely on its own.

- **Spyware**

- **What it is:** Software that secretly hides on your computer and monitors your activity without your consent.
- **Study Tip:** Spyware acts exactly like a digital spy. It can record your keystrokes to steal your passwords, track your browsing history, and quietly send your personal data back to cybercriminals.

- **Antivirus**

- **What it is:** A critical piece of utility software designed to scan for, detect, block, and safely remove malware from your system.
- **Study Tip:** Think of it as your computer's immune system or a virtual security guard. It constantly scans incoming files and compares them against a massive database of known digital threats.

- **Cyber Safety**

- **What it is:** The general rules, behaviors, and everyday best practices you should follow to protect yourself online.
- **Study Tip:** This includes proactive actions like using strong and unique passwords, not clicking on suspicious email links or attachments (phishing), avoiding public Wi-Fi for banking, and keeping your Operating System updated to patch security holes.